

EXPERIMENT - 6

TO DETERMINE THE SOLUBILITY OF THE GIVEN WATER SOLUBLE SOLID AT ROOM TEMPERATURE.

APPARATUS REQUIRED

- | | |
|---------------------------|-----------------------|
| a) <u>Beaker</u> | b) <u>Glass rod</u> |
| c) <u>Filter Paper</u> | d) <u>Funnel</u> |
| e) <u>Porcelain basin</u> | f) <u>Stand</u> |
| g) <u>Wire gauze</u> | h) <u>Thermometer</u> |

CHEMICAL REQUIRED

- | | |
|---------------------------------|------------------|
| i. <u>Given salt (i.e NaCl)</u> | ii. <u>Water</u> |
|---------------------------------|------------------|

THEORY

The solubility of a solute in a given solvent varies appreciably with temperature. In general, solubility of substance in a solvent increases with increase in temperature but solubility of few substances decrease with increase in temperature. Hence solubility of a solute is affected by rise or fall in temperature. Thus the determination of solubility of a solute at laboratory temperature involves the preparation of certain amount of saturated solution followed by determination of amount of solute present in the known weight of saturated solution by evaporation of solvent completely. The weight of solute and solvent can then be determined and put in the solubility formula to find out the solubility.

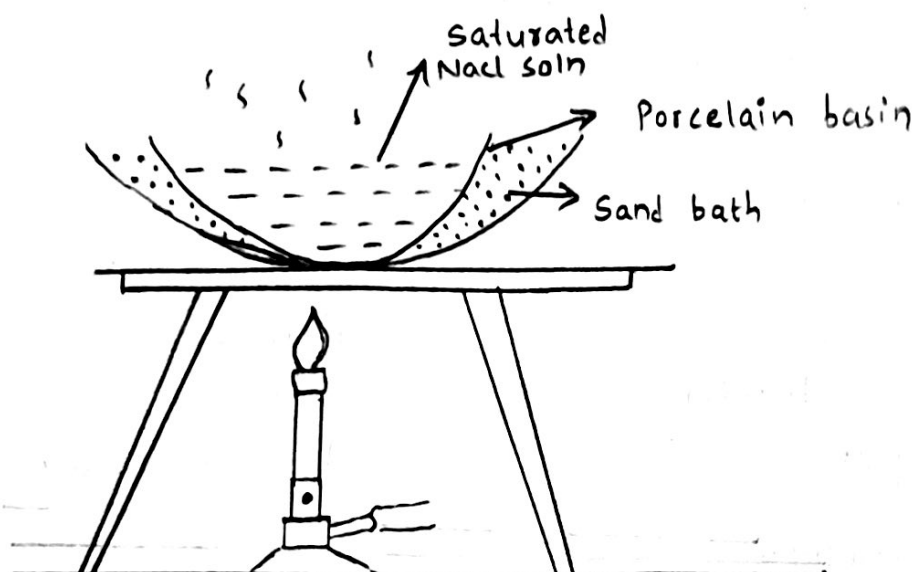
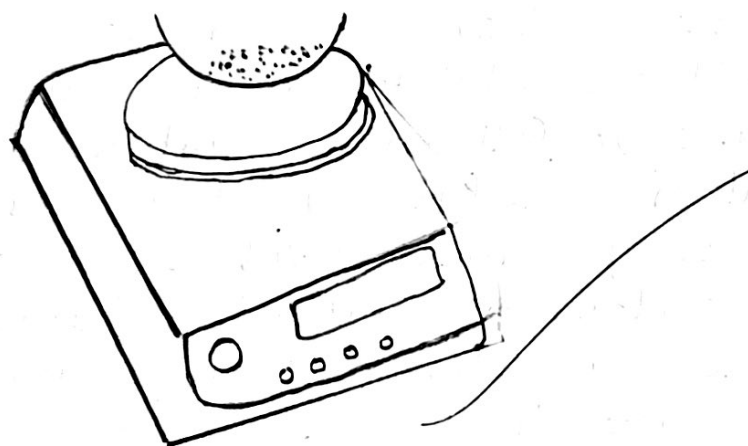


fig - Evaporation



Taking the weight of NaCl soln

of solute at room temperature.

Solubility of a substance is defined as the maximum amount of solute in gram that can be dissolved in 100 gram of a solvent to make saturated solution at a given temperature.

Mathematically,

$$\text{Solubility} = \frac{\text{Amount of solute (in gram)}}{\text{Amount of solvent (in gram)}} \times 100$$

PROCEDURE

20ml of saturated solution of NaCl was prepared in distilled water in a beaker at laboratory temperature. The solution was made in such way that small amount of NaCl ultimately remained undissolved. The empty porcelain basin was weighed. The saturated solution of NaCl and water was filtered and filtrate was transferred in porcelain basin. The porcelain basin with saturated solution was weighed. The solution was heated and evaporated up to dryness over sand bath, burner. The porcelain basin with solid residue was weighed again. The temperature of Laboratory was measured. All the other

Necessary things were also noted.

OBSERVATION

Laboratory temperature = 21°C

Weight of empty porcelain basin = 66.4 gm

Weight of porcelain basin and soln = 105.02 gm

Weight of ~~soln~~ porcelain basin and solute
= 77.82 gm

CALCULATION

$$\text{Weight of solute} = (W_2 - W_1) \text{ gm} = 77.82 - 66.4 \\ = 11.42 \text{ gm}$$

$$\text{Weight of Solvent} = (W_2 - W_3) \text{ gm} = 105.02 - 77.82 \\ = 27.2$$

Now,

$$\text{Solubility} = \frac{\text{Weight of Solute}}{\text{Weight of Solvent}} \times 100$$

$$= \frac{(W_2 - W_1)}{(W_2 - W_3)} \times 100 = \frac{11.42}{27.2} \times 100$$

$$= 41.98 \%$$

RESULT

In this way the solubility of the given salt at laboratory temperature was found to be 41.98

CONCLUSION

Hence From above experiment it is concluded that the solubilities of various water soluble substances can be determined at laboratory temperature.

PRECAUTIONS

- ① Personal protective equipments should be worn.
- ② The solution should be saturated.
- ③ Evaporation process should be gentle and should not be overheated.
- ④ Heating, cooling and weighing should be done carefully.
- ⑤ Chemical balance should be handled properly.

VIVA-VOCE

1. How does temperature affect solubility?
2. What is solubility curve?
3. What kind of solute gives discontinuous solubility curve?
4. What is saturated solution?
5. What is transition temperature?

① Solubility of most of the salts increase with increase in temperature but solubility of some salts decrease with increase in temperature.

② The curve obtained by plotting solubilities of various substances at different temperatures.

③ The salt having water of crystallization gives discontinuous solubility curve (i.e. hydrated salt.)

④ The solution which cannot dissolve any more solutes at room temperature.

⑤ The temperature at which a substance changes from one phase to another phase, For example: transition temperature of sodium sulphate is 32.4°C at which hydrated phase changes to anhydrous phase.

Meer
9/9